

Manguilesy Corentin

Procédure – Réplication Maître ↔ Maître MariaDB

Objectif

Mettre en place une réplication bidirectionnelle entre deux serveurs MariaDB pour la base 'gsb_frais'.

Configuration des serveurs

Élément	intralabcm	hdintralabcm
IP	172.18.XX	172.18.X.X+1
Server ID	1	2
bind-address	172.18.XX	172.18.X.X+1
auto_increment_offset	1	2

Étapes à suivre

1. Configurer MariaDB

Fichier : /etc/mysql/mariadb.conf.d/50-server.cnf

- À ajouter/modifier dans [mysqld] :
- server-id = X # 1 ou 2 selon le serveur
- log_bin = binlog
- relay_log = relay-bin
- log_slave_updates = 1
- auto_increment_increment = 2
- auto_increment_offset = X # 1 ou 2 selon le serveur
- expire_logs_days = 10
- max_binlog_size = 100M
- replicate-do-db = gsb_frais
- bind-address = IP_du_serveur

Redémarrer MariaDB :

sudo systemctl restart mariadb

2. Créer l'utilisateur 'replica'

- À exécuter sur chaque serveur :

```
CREATE USER 'replica'@'%' IDENTIFIED BY 'replica_password';  
GRANT REPLICATION SLAVE ON *.* TO 'replica'@'%';  
FLUSH PRIVILEGES;
```

3. 📖 Obtenir les infos binlog

- Sur chaque serveur :

```
SHOW MASTER STATUS;
```

Notez : File (nom du binlog) et Position (position du journal).

4. 📖 Configurer la réplication croisée

- Sur intralabcm :

```
CHANGE MASTER TO  
MASTER_HOST='172.18.X.X+1',  
MASTER_USER='replica',  
MASTER_PASSWORD='replica_password',  
MASTER_LOG_FILE='binlog.000001',  
MASTER_LOG_POS=XXX;  
START SLAVE;
```

- Sur hdintralabcm :

```
CHANGE MASTER TO MASTER_HOST='172.18.X.X',  
MASTER_USER='replica',  
MASTER_PASSWORD='replica_password',  
MASTER_LOG_FILE='binlog.00000X',  
MASTER_LOG_POS=XXX;  
START SLAVE;
```

5. ✔ Vérifier la réplication

- Sur chaque serveur :

SHOW SLAVE STATUS\G

Vérifier que : Slave_IO_Running: Yes et Slave_SQL_Running: Yes

Résultat sur mes deux serveurs:

Fichier : /etc/mysql/mariadb.conf.d/50-server.cnf

Intralabcm:

```
GNU nano 5.4 /etc/mysql/mariadb.conf.d/50-server.cnf
#
# These groups are read by MariaDB server.
# Use it for options that only the server (but not clients) should see
#
# this is read by the standalone daemon and embedded servers
[server]
#
# this is only for the mysqld standalone daemon
[mysqld]
server-id=1
log_bin=binlog
relay_log=relay-bin
log_slave_updates=1
auto_increment_increment=2
auto_increment_offset=1
expire_logs_days=10
max_binlog_size=100M
master-retry-count=20
replicate-do-db=gsb_frais

bind-address = 172.18.
#
# * Basic Settings
#
user                        = mysql
pid-file                   = /run/mysqld/mysqld.pid
basedir                   = /usr
datadir                   = /var/lib/mysql
tmpdir                    = /tmp
lc-messages-dir           = /usr/share/mysql
lc-messages               = en_US
skip-external-locking

# Broken reverse DNS slows down connections considerably and name resolve is
# safe to skip if there are no "host by domain name" access grants
#skip-name-resolve

# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address              = 172.18.
#
# * Fine Tuning
```

Résultat de la commande SHOW SLAVE STATUS\G

```
MariaDB [(none)]> SHOW SLAVE STATUS\G
***** 1. row *****
      Slave_IO_State: Waiting for master to send event
        Master_Host: 172.18.
        Master_User: replica
        Master_Port: 3306
        Connect_Retry: 60
        Master_Log_File: binlog.000003
      Read_Master_Log_Pos: 339
        Relay_Log_File: relay-bin.000010
        Relay_Log_Pos: 635
      Relay_Master_Log_File: binlog.000003
      Slave_IO_Running: Yes
      Slave_SQL_Running: Yes
      Replicate_Do_DB: gsb_frais
      Replicate_Ignore_DB:
      Replicate_Do_Table:
      Replicate_Ignore_Table:
      Replicate_Wild_Do_Table:
      Replicate_Wild_Ignore_Table:
        Last_Errno: 0
        Last_Error:
        Skip_Counter: 0
      Exec_Master_Log_Pos: 339
      Relay_Log_Space: 1234
      Until_Condition: None
      Until_Log_File:
      Until_Log_Pos: 0
      Master_SSL_Allowed: No
      Master_SSL_CA_File:
      Master_SSL_CA_Path:
      Master_SSL_Cert:
      Master_SSL_Cipher:
      Master_SSL_Key:
      Seconds_Behind_Master: 0
Master_SSL_Verify_Server_Cert: No
        Last_IO_Errno: 0
        Last_IO_Error:
        Last_SQL_Errno: 0
        Last_SQL_Error:
      Replicate_Ignore_Server_Ids:
      Master_Server_Id: 2
      Master_SSL_Crl:
      Master_SSL_Crlpath:
        Using_Gtid: No
        Gtid_IO_Pos:
      Replicate_Do_Domain_Ids:
      Replicate_Ignore_Domain_Ids:
        Parallel_Mode: optimistic
        SQL_Delay: 0
        SQL_Remaining_Delay: NULL
      Slave_SQL_Running_State: Slave has read all relay log; waiting for more updates
      Slave_DDL_Groups: 0
      Slave_Non_Transactional_Groups: 0
      Slave_Transactional_Groups: 0
1 row in set (0,000 sec)
```

Hdintralab:

```
#
# These groups are read by MariaDB server.
# Use it for options that only the server (but not clients) should see

# this is read by the standalone daemon and embedded servers
[server]

# this is only for the mysqld standalone daemon
[mysqld]
server-id=2
log_bin=binlog
relay_log=relay-bin
log_slave_updates=1
auto_increment_increment=2
auto_increment_offset=2
expire_logs_days=10
max_binlog_size=100M
master-retry-count=20
replicate-do-db=gsb_frais

bind-address = 172.18.

#
# * Basic Settings
#

user                = mysql
pid-file             = /run/mysqld/mysqld.pid
basedir              = /usr
datadir              = /var/lib/mysql
tmpdir               = /tmp
lc-messages-dir      = /usr/share/mysql
lc-messages          = en_US
skip-external-locking

# Broken reverse DNS slows down connections considerably and name resolve is
# safe to skip if there are no "host by domain name" access grants
#skip-name-resolve

# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address         = 172.18.
```

Résultat de la commande SHOW SLAVE STATUS\G:

```
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW SLAVE STATUS\G
***** 1. row *****
      Slave_IO_State: Waiting for master to send event
        Master_Host: 172.18.1.1
        Master_User: replica
        Master_Port: 3306
        Connect_Retry: 60
        Master_Log_File: binlog.000013
      Read_Master_Log_Pos: 339
        Relay_Log_File: relay-bin.000011
        Relay_Log_Pos: 552
      Relay_Master_Log_File: binlog.000013
      Slave_IO_Running: Yes
      Slave_SQL_Running: Yes
      Replicate_Do_DB: gsb_frais
      Replicate_Ignore_DB:
      Replicate_Do_Table:
      Replicate_Ignore_Table:
      Replicate_Wild_Do_Table:
      Replicate_Wild_Ignore_Table:
        Last_Errno: 0
        Last_Error:
        Skip_Counter: 0
      Exec_Master_Log_Pos: 339
      Relay_Log_Space: 1234
      Until_Condition: None
      Until_Log_File:
      Until_Log_Pos: 0
      Master_SSL_Allowed: No
      Master_SSL_CA_File:
      Master_SSL_CA_Path:
      Master_SSL_Cert:
      Master_SSL_Cipher:
      Master_SSL_Key:
    Seconds_Behind_Master: 0
Master_SSL_Verify_Server_Cert: No
      Last_IO_Errno: 0
      Last_IO_Error:
      Last_SQL_Errno: 0
      Last_SQL_Error:
  Replicate_Ignore_Server_Ids:
      Master_Server_Id: 1
      Master_SSL_Crl:
      Master_SSL_Crlpath:
        Using_Gtid: No
        Gtid_IO_Pos:
      Replicate_Do_Domain_Ids:
      Replicate_Ignore_Domain_Ids:
        Parallel_Mode: optimistic
        SQL_Delay: 0
      SQL_Remaining_Delay: NULL
      Slave_SQL_Running_State: Slave has read all relay log; waiting for more updates
      Slave_DDL_Groups: 0
  Slave_Non_Transactional_Groups: 0
      Slave_Transactional_Groups: 0
1 row in set (0.000 sec)
```

Test de la relation maître-maître :

Saisie d'une fiche de fraie via l'application sur le serveur intralabcm :

Renseigner ma fiche de frais du mois de Avril 2025

Les modifications de la fiche de frais ont bien été enregistrées

Eléments forfaitisés

* Forfait Etape :

0

* Frais Kilométrique :

0

* Nuitée Hôtel :

0

* Repas Restaurant :

0

Valider

Effacer

Descriptif des éléments hors forfait

Date	Libellé	Montant	
2025-04-18	TEST	1233.00	Supprimer

Nouvel élément hors forfait

* Date :

18/04/2025

* Libellé :

TEST

* Montant :

1233

Ajouter

Effacer

La fiche a été créer :

Fiche de frais du mois de Avril 2025 : *Fiche créée, saisie en cours* depuis le 2025-04-18

Montant validé :

Quantités des éléments forfaitisés

Forfait Etape	Frais Kilométrique	Nuitée Hôtel	Repas Restaurant
0	0	0	0

Descriptif des éléments hors forfait - 0 justificatifs reçus -

Date	Libellé	Montant
2025-04-18	TEST	1233.00

Arrêt du noeud intralabcm:

```
crm(live/intralabcm)# node standby intralabcm
crm(live/intralabcm)# status
Cluster Summary:
* Stack: corosync
* Current DC: intralabCM (version 2.0.5-ba59be7122) - partition with quorum
* Last updated: Fri Apr 18 20:18:06 2025
* Last change: Fri Apr 18 20:18:01 2025 by root via crm_attribute on intralabCM
* 2 nodes configured
* 2 resource instances configured

Node List:
* Node intralabCM: standby
* Online: [ hdintralabCM ]

Full List of Resources:
* Resource Group: servIntralab:
* IPFailover (ocf::heartbeat:IPaddr2): Started hdintralabCM
* serviceWeb (systemd:apache2): Started hdintralabCM

crm(live/intralabcm)#
```

Lors de la reconnexion, on constate que les données sont toujours présentes et identiques malgré l'arrêt du nœud intralabcm, ce qui confirme le bon fonctionnement de la réplication entre les serveurs et, par conséquent, de la haute disponibilité.

Mes fiches de frais

Mois à sélectionner :

Mois : Avril 2025 ▼

Valider Effacer

Fiche de frais du mois de Avril 2025 : *Fiche créée, saisie en cours depuis le 2025-04-18*

Montant validé :

Quantités des éléments forfaitisés

Forfait Etape	Frais Kilométrique	Nuitée Hôtel	Repas Restaurant
0	0	0	0

Descriptif des éléments hors forfait - 0 justificatifs reçus -

Date	Libellé	Montant
2025-04-18	TEST	1233.00
2025-04-18	Test après arrêt intralab	1.00

Mise en marche du noeud :

```
sio@intralabcm:~$ sudo crm
crm(live/intralabcm)# node online intralabcm
crm(live/intralabcm)# status
Cluster Summary:
* Stack: corosync
* Current DC: intralabCM (version 2.0.5-ba59be7122) - partition with quorum
* Last updated: Fri Apr 18 20:22:44 2025
* Last change: Fri Apr 18 20:22:41 2025 by root via crm_attribute on intralabCM
* 2 nodes configured
* 2 resource instances configured

Node List:
* Online: [ hdintralabCM intralabCM ]

Full List of Resources:
* Resource Group: servIntralab:
  * IPFailover (ocf::heartbeat:IPaddr2): Started intralabCM
  * serviceWeb (systemd:apache2): Starting intralabCM
```